UNIT 4: Linear models

Some statistical tests

- anova, t-test, Linear regression
- chi square, F test

Linear models of observations

- * t test
  - * including paired t, one-sample t
- * regression
  - * including Pearson correlation
- * anova
- * ancova

Outline

Approach and assumptions

Examples
- Assumption checking
- Other considerations

More complicated models
- Multi-parameter variables
- Multiple predictor variables
- Interactions

What is a linear model?

- A *fit* to a **response variable** based on assumptions about its relationship to one or more **predictor variables**
- *Inferences* about parameters based on assumptions about the *conditional* distribution of the response variable
  - Assume the response variable has "true" distribution for any value of the predictor variables
    - Use temperature to predict deaths
- A linear model gives the response variable as a linear combination of the *parameters* it is estimating
  - * It is not necessarily linear with respect to predictor variables!

Linear model?

y is response, x is predictor

- $y = a + bx$
- $y = a + bx + cx^2 + dx^3$
- $y = \frac{a}{b+x}$
- $y = ax^b$
- $\log(y) = \log a + b \log(x)$
- * 1, 2 and 5 are lms
  - * Linear in a, b, ...
- * 4 and 5 are the same equation, but imply a different *distribution* for the response variable y

Linear model?

- What if x is a categorical variable?
  - * *Any* response to a categorical variable can be encoded as a linear model!
- *What do we mean by linear in the parameters?*
  - *Consists of "terms" that are added together*
  - *Each term is multiplied by a single parameter, one time*

Power

- Linear models are very efficient
  - They find the best "least-squares" fit with simple linear algebra
- They can be very powerful at detecting effects
  - They can also be over-sensitive to outliers
  - *Spiderman: with great power comes great responsibility*



Assumptions

- Linear-model *fits* are basically always correct, but can be hard to interpret
  - You can find the "best" parameters
  - But you need assumptions to know what that means
- Linear-model *inferences* (parameter estimates P-values, CIs) depend on assumptions about *residuals*
- A residual is $y - \hat{y}$: the difference between an observed value y and a model prediction $\hat{y}$
- lm inference assumes that residuals are independent and normally distributed, with constant variance
  - The mean of the residuals is always 0
  - * Otherwise the machine would find a better linear model

Variables

- What kind of variables can we use?
- Response
  - * We assume the response variable is normally distributed (conditional on the predictor variables)
  - * Therefore it must be continuous
- Predictors
  - * The linear model framework gives us a lot of flexibility to construct a model
  - * Predictor variables can be of any type
  - * e.g., Continuous, integer, ordinal, nominal

Fitting in R

- $\mathrm{lm}(y \sim 1 + x + c)$
  - Use y as a response variable, x and c as predictors
  - 1 refers to the *intercept*, which is a constant term
- $\mathrm{lm}(y \sim x + c)$
  - * *Also* has an intercept
  - * intercepts are usually good
  - * $\mathrm{lm}(y \sim 0 + x + c)$ to suppress it
    - * *or* $\mathrm{lm}(y \sim x + c - 1)$ to suppress it

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No predictor

- **One-sample t-test**
  - e.g., what is the mean weight of an African elephant?
- Assumption: population is normally distributed
- Example: fev

Check the assumption

- * Is the sample normally distributed?
  - * No! Nothing in biology is perfect
- * Does it look close to normal?
  - * Should you consider another test, or another question?
    - * Permutation test of the mean
    - * Test the median instead
    - * Test the mean on a transformed scale

Interpreting diagnostics

- Your linear model assumptions are not met exactly
  - You can't use failure to reject the null hypothesis as evidence that your assumptions are OK
- There is no universally accepted way to decide whether you are close enough!
  - When sample size gets larger, your ability to reject the null hypothesis get stronger
  - But concerns often get smaller
    - Particularly normality concerns
    - * central limit theorem!
- We may circle back to this for projects

Interpreting results

- Generally, avoid R's summary table!
  - * It shows P values but not CIs
- Focus on estimates and confidence intervals
- But how to interpret these?
  - * Pay attention to units
  - * You need an outside reference
  - * Statistics is not magic

Binary predictor

- Two-group test
  - Predictor with two values (could be continuous or categorical)
  - Can men hold their breath longer than women?
  - Are mice more active at 10 degrees or 20 degrees C?
  - Generally interested in:
    - * The sign of the difference
    - * Also the magnitude (size)
  - *Can we clearly see the sign of the difference?*
    - P value
  - *How big might the difference be*
    - Confidence interval
  - *Can we clearly see the sign of the difference?*
    - P value
- Example mouseActivity

Testing assumptions

- * Residuals should look approximately normal
- * and approximately the same across the two groups
- Otherwise? Same answers as above
  - * Permutation test
  - * Different scale
  - * Medians

R model parameters

- Intercept (constant) is the estimated value at:
  - zero for continuous predictors
  - "baseline" for categorical predictors
- Other parameters
  - slopes for continuous predictors
  - differences for categorical predictors

Effect sizes

- What are the units of our parameter in this example?
  - * ml CO2/h for the mouse activity
    - * Or: (ml CO2/h)/degC!
  - * inches for the height; L for the volume.
- How should we know if they are large or not?
  - * Compare to population statistics, not inferential statistics
  - * We want to know if the value is large compared to something real, not compared to our own uncertainty!
  - * e.g., total value, variation in population, difference between sexes, difference between species . . .
  - * Use a log scale!
  - Specific scientific knowledge may be needed
  - Compare to something with the same units

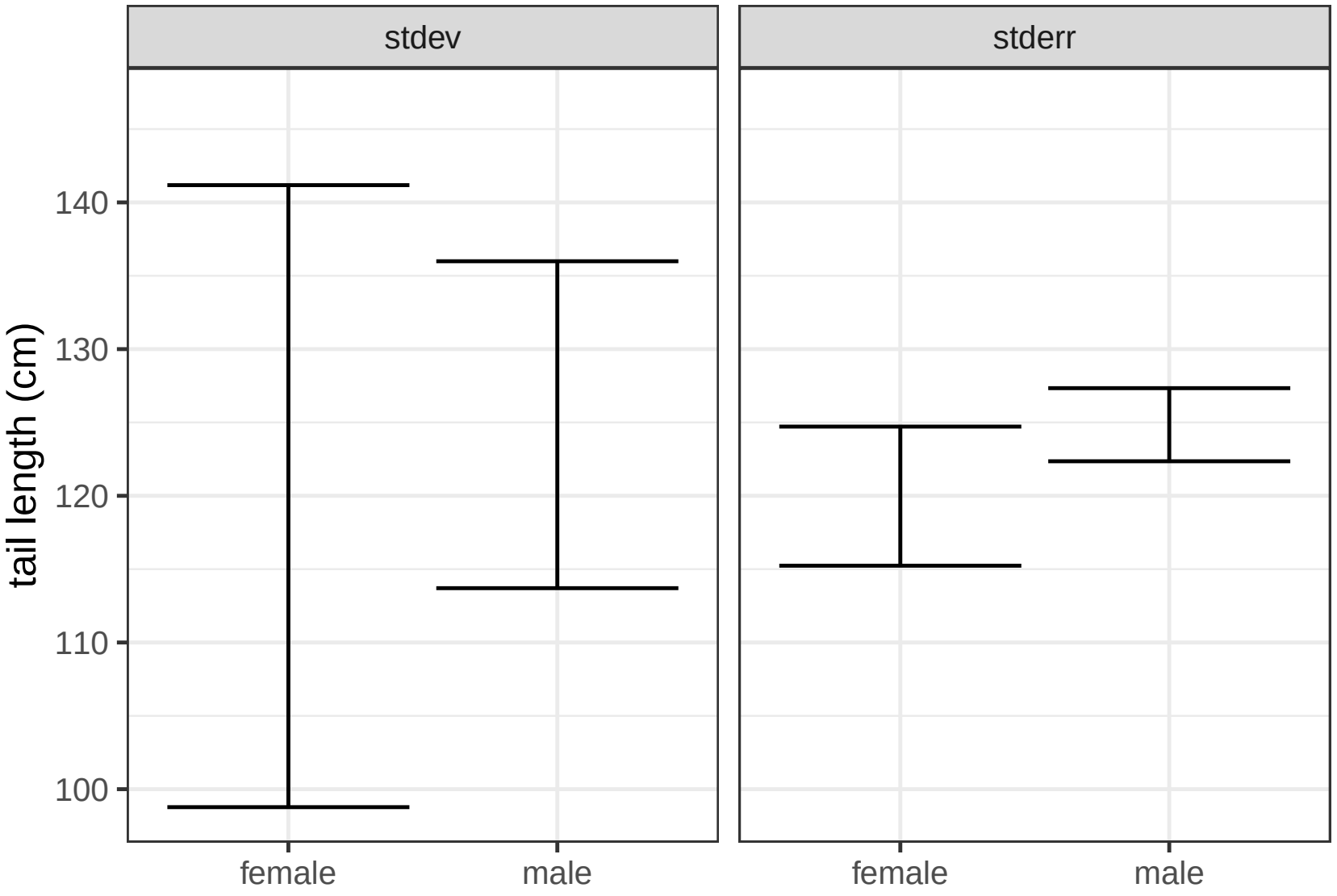
Measures of size

- We can examine size of a parameter in two ways
- Certainty
  - * t value, P value
  - * This is usually our certainty about the sign of something – but what?
- Effect
  - * Unit-ed value, ratio of value to something with same units
  - * or a log ratio
- *Ratios are also good (even though JD has a log bias)*

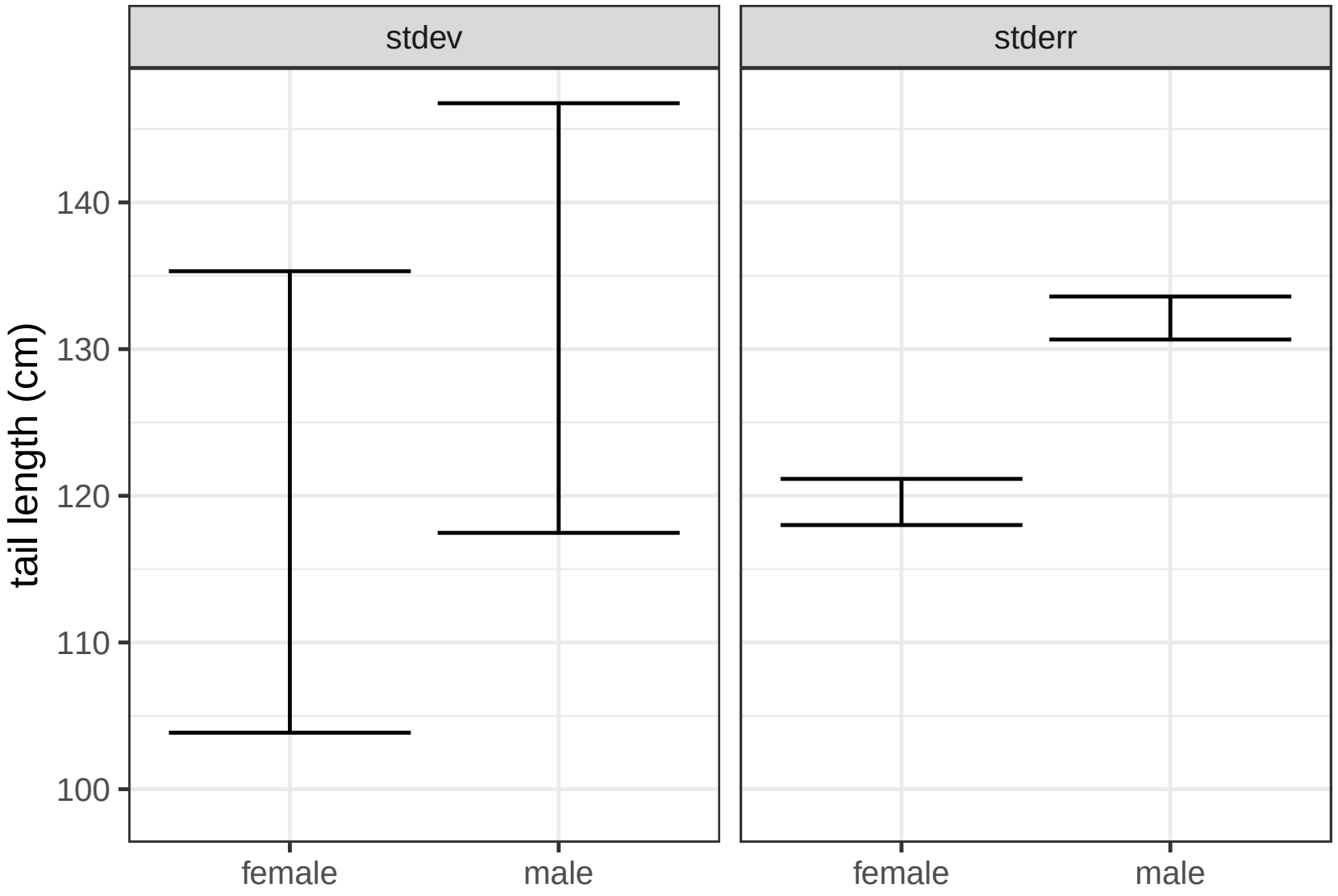
Measures of spread

- Standard deviation
  - * How much variation we see in the population
  - * Compare this to understand the effect estimate
- Standard error (or confidence interval)
  - * How much variation we would expect to see if estimated the parameter again
  - * Compare this to understand the certainty of something

Small sample



Large sample



More on effect size

- Scientifically, are you more interested in knowing:
  - How big an effect is, or how certain you are about its sign?
  - * Usually how big the effect is
    - *Certainty is important, but we don't usually focus on measuring how certain we are*
- Why is there so much emphasis on . . .
  - * P values?
  - * Some people are afraid of statistics and think it's magic
  - * Some people want statistics to be magic
    - * e.g., so that they don't have to make decisions about effect sizes
    - * So they can say things "don't matter"

lm Questions

- t-test vs. anova vs. regression
  - make sure your predictor variables are coded correctly, and then don't worry about it
- different-sized groups
  - don't worry about it
- *lm is kind-of magic*

Standard deviation

- Measures variability of a *variable*
  - Same units as the variable
  - * Variance is the square; mathematically awesome, biologically confusing
- Estimate divided by within-group standard deviation can help understand biological importance
  - * E.g., the effect of the temperature treatment is 1.2$\times$ the amount of natural variability
  - Often called d or g: fields have cutoff numbers for this, but JD thinks it's better to use scientific judgment usually

Standard error

- Measures uncertainty of a *parameter*
  - Same units as the parameter
  - * The same as the variable for a categorical predictor
  - *Units of reponse per units of predictor for a continuous predictor*
- Estimate divided by standard error is a measure of certainty
  - Directly related to P value
  - Often reported (in R's summary as the t statistic)
    - We won't use it here
    - ADD https://xkcd.com/1378/

Parameterizing

- Different two-group models have the same fit
  - * predicted mean for each observation
  - * estimated standard deviation for each observation
  - *P value for the sign of the difference*
- but different parameterizations
  - baseline and difference (default for categorical variables)
  - intercept and slope (default for continuous variables)
  - mean and difference
- lm does not work with too many parameters
  - * You would need to choose two in this case
  - * Because we're interested in a mean for each group

Paired t-test

- Example: do people have more joint erosion on their dominant hand?
  - * Can't treat my left hand and right hand as independent!
- How to do it?
  - * The paired t-test is equivalent to a one-sample test on the *differences*!
  - * Procedure is the same, but now we're particularly interested in the *sign* of the estimate
- Example bengaliDigits
- CLASS See notes for data link

Continuous predictor

- Regression analysis
- Typically focus on the slope
- Pay attention to baseline as well as intercept
- Example: fev.R

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Assumption checking

- We want to think about:
  - Independence
  - Linearity
  - Homogeneity of variance
  - Normality
- Roughly in that order

Independence

- The most important assumption. Can't really be checked statistically
- Good study design, good model construction
- Observations should not share anything with other observations that's not included in the model
  - They should be conditionally independent

Independence example

- What should we worry about in the lung example?
  - * is the same child measured twice?
  - * are some groups of children related?
  - * Are older children taken from different schools?
- Does it depend on our scientific question?
  - * Do we want to know if height is *driving* lung capacity?
    - * NOANS
  - * Or just if is *associated*?

Study design

- Randomized studies, if done correctly, can address most independence concerns
- But not all studies can be randomized
  - * NOANS

Linearity

- Kind of hard to state:
  - Is a model that is linear in these particular parameters a good fit to the data?
- If not:
  - Consider whether you should transform
  - Or add curvy terms (polynomial or spline)
  - Or if you need interactions

Homogeneity of variance

- * Opposite is heterogeneity of variance
- *lm inference assumes that variance does not change with the predictor variables*
- *also called homo- and heteroscedasticiy*
- When would you expect to see heteroscedasticiy?
  - * If you have a large range of positive predictions
  - *If variation is proportional to size, log transformations can help ...*
- Consider a data transformation

Normality

- Least important consideration
  - And the one that is most helped by the Central Limit Theorem
- Can often be addressed (if necessary) by modifying the linear model
  - robust lm inference
  - permutation-based lm inference

The Central Limit Theorem

- Most distributions converge to a normal distribution if you add up independent random variables and take the mean
  - Variables that result from additive processes will tend to be normal
  - *Variables that result from multiplicative processes will tend to be lognormal*
  - Tests based on normality assumptions will be less sensitive to that assumption when there is more data
    - There is adding and averaging going on inside the test

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Testing the regression assumptions

- Be careful about testing with P values
  - * Assumptions are never exactly true
  - * We will rarely reject when sample size is small, and often reject when sample size is large. How does that help?
- Look for patterns in your diagnostic plots, and think about options

Predictive check

- Use `check_model` from the performance package
- General feeling about all the assumptions
- Simulates what your distribution would look like if the assumptions were true
- Compares to actual data

Outliers

- Is there a single point with a very large influence on your whole conclusion?
- How do you feel about that?
- Sensitivity analysis
  - * Try the analysis without some combination of outliers just to see
- Try to rely less on normal assumption
  - * robust or permutation-based inference

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What if you don't think response is a straight line?

- Transform the response
- Transform the predictor
- Fit a curve
- These can also mean: changing your scientific question to some extent
  - What is the slope on the linear scale?
  - What is the slope on the log scale?
  - What is the average slope across a curved response?

Categorical predictor

- Classically called an anova
- We *add* $n-1$ new parameters for a variable with n categories
  - We also add $n-1$ "lm variables" to the calculation
  - These are often called "dummy variables"
  - They code which category each observation is in, using 0s and 1s
- Example chickWeights

What paradigm to use?

- Even assuming we are frequentist
- We need to fit $n-1$ parameters; how do we deal with α?
  - * The *counterfactual* probability of seeing something when there is nothing there
- At what *level* should we control α?
  - * Experiment
  - * Variable
  - * Parameter

Hypothesis-based analysis

- Best is to focus on a small set of specific questions
- Associate them with a small number of parameters you can measure
  - You can use contrasts or interactions to construct the test you are most interested in

Exploratory analysis

- Look at a large number of effects
- Which ones seem interesting?
- Should you believe them?
  - * No
  - * Maybe make some hypotheses and construct a hypothesis-based analysis
  - *Then: do a new experiment*
- False discovery rates and P value corrections should be used sparingly
- Can also look at confidence intervals
  - *Be aware that some proportion do not contain the answer*

Variables and parameters

- To fit a quadratic, we need two parameters for the response
- This also means we need two "lm variables" for the model
  - * x and x^2
- We might want to fit other sorts of curves (typically a spline)
- Like feed (in the chicks) this is a "multi-parameter variable"
  - * Specify your frequentist standard before testing

Fitting a quadratic

- Example fev
- Do we have a specific scientific question?
  - Does it curve up or down?
    - * Test the quadratic term
  - Can we be confident about the average trend, in the presence of curvature?
    - * Look at CIs for the linear term
- Use R function poly (see example)
  - Beware of overfitting

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Multiple predictor variables

- Linear models can easily handle multiple predictor variables
- Example: fev
- Depending on our scientific question, more variables can take us closer or farther from the answer we want

Scientific question

- Height can mediate the relationship between age and lung capacity
  - * People might be taller because they're older
  - * We might want to exclude it to see the effects of age
  - * Or include it to see total effects
- Age cannot be caused by height
  - * We should (usually) include it to see the effect of height

Controlling

- Adding a second variable when our primary interest is the first is called "controlling"
  - * We control for age when we study the effects of smoking on lung capacity
  - * Also the effect of sex
- When you control for a variable, you don't necessarily need to interpret the parameters of that variable at all!

Conflation

- Why do children who smoke have higher lung capacity?
  - * Because they're older
  - * Age causes both of these things, separately from any effect they have on each other
- This is called conflation, or confounding

Accounting for variation

- Even when we don't have conflation, controlling for relevant predictors can make our predictions more accurate
- Example, randomized controlled trial of a blood-pressure drug
  - * No conflation – the only thing that effects our main predictor is the coin!
  - * Still want to control for things

How many parameters?

- Statistics isn't magic
  - The linear model will find patterns even in noise
- Controlling for too many factors will make our predictions less *precise*
  - * Add variation, uncertaint
- Rule of thumb:
  - 5–10 solid, continuous data points per parameter
  - When data points are small counts, or yes/no, we need to reduce parameters further

Which parameters?

- It's very common to choose predictors based on preliminary analyses
- Or to throw out interaction or curvature terms based on significance
- This is accepting the null hypothesis (concluding these things don't matter)
- It's also introducing confirmation bias
  - If we choose our final test based on the data, we will be bad frequentists
  - * wrong more than 5% of the time

Which parameters?

- Decide how many parameters you can afford based on the data
- Decide which parameters to use based on scientific judgment
  - Can be based on other data, but not based on data from your study

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What is an interaction?

- So far, each predictor has its own effect, and the effects add up
  - The effect of age on lung capacity is assumed to be the same for boys and girls
- An **interaction** means that the effect of one predictor depends on the value of another
  - Does lung capacity increase faster with age for boys than for girls?
- Interactions are symmetric
  - * If the effect of age depends on sex, then the effect of sex depends on age
  - * We can choose whichever way of saying it matches our question

Interactions in R

- lm($y \sim x + z$)
  - Additive model: no interaction
- lm($y \sim x * z$)
  - Short for lm($y \sim x + z + x : z$)
  - The new "lm variable" is made by multiplying x and z
- Is this still a linear model?
  - * Yes! It's linear in the *parameters*
- How many parameters does an interaction add?
  - * Multiply the parameters of the two variables
  - * Continuous and two categories: 1
  - * Three categories and four categories: $2 \times 3 = 6$
- Example classPenguins (based on lmAsn)

Interpreting parameters

- $y = a + bx + cz + dxz$, where z is 0 for baseline, and 1 for the other group
  - Baseline group: intercept is a, slope is b
  - * Other group: intercept is $a + c$, slope is $b + d$
- What does d mean?
  - * The *difference* between the slopes
- What does c mean now?
  - * The difference between the groups when x is zero
  - * This may not be interesting, or even make sense (height zero?)
  - * Centering x can make it mean something
- Parameters for the other variables are no longer overall effects

Main effects

- Your model is now estimating many different possible effects for one predictor
  - * effect is different when other predictors change
- Depending on your scientific question, you can:
  - Ignore these
  - Average over them (may return to that)
- This confusion is often used as an excuse to throw out non-significant interactions
  - * Do not do that!

Is one group changing more than another?

- Treatment group changes significantly; control group does not
  - Can we say that treatment group changed more?
  - * No! Significant and not significant are not (necessarily) significantly different
  - * Failing to reject for the control group doesn't mean it didn't change
- The interaction tests the question directly
  - * Group $\times$ time: how different are the changes?
  - * The estimate and CI are for the difference we actually care about

Interactions and scale

- If effects combine by multiplying, they look like an interaction on the linear scale
  - * A treatment that increases growth by 10% adds more grams to big animals
  - * On the log scale, there may be no interaction
- Whether there's an interaction can depend on the scale you choose
  - * Decide what scale makes scientific sense before interpreting
  - * This is why transformations and interactions both came up under linearity

Which interactions?

- A non-significant interaction doesn't mean that there is no interaction
  - * Throwing it out is accepting the null hypothesis
- Same answer as for other parameters:
  - * Decide how many you can afford, and choose based on scientific judgment
- It's often good to boil things down to a one-parameter interaction
- Beyond that, things can get complicated.

Example: comparing to the control group

- We test reflex times in two groups of rabbits
  - One exposed to radiation, and one not
    - * usually the non-radiation group is a sham, where you pretend to expose them to radiation
- Naive method: show that the sham group doesn't change, and the radiation group does
  - * Better method: show that the sham group changes *differently* (hypothesis is that they will be *relatively* slower)
- Typically done with interactions
  - Although sometimes it could be done more simply with differences